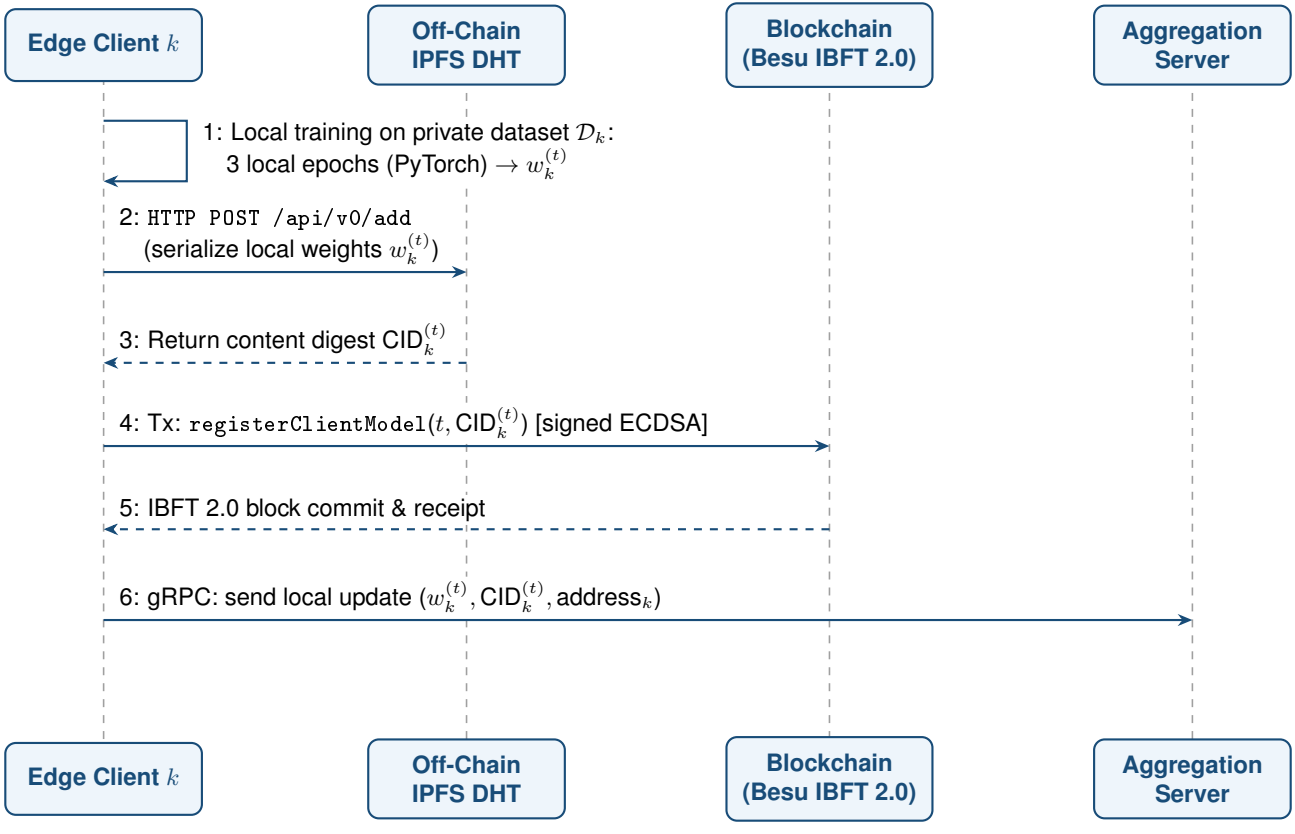

Edge Client k
Off-Chain IPFS DHT
Blockchain (Besu IBFT 2.0)
Aggregation Server
1: Local training on private dataset $\mathcal{D}_k$:
3 local epochs (PyTorch) $\rightarrow w_k^{(t)}$
2: HTTP POST /api/v0/add
(serialize local weights $w_k^{(t)}$)
3: Return content digest $\text{CID}_k^{(t)}$
4: Tx: registerClientModel$(t, \text{CID}_k^{(t)})$ [signed ECDSA]
5: IBFT 2.0 block commit & receipt
6: gRPC: send local update $(w_k^{(t)}, \text{CID}_k^{(t)}, \text{address}_k)$
Edge Client k
Off-Chain IPFS DHT
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